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# Doping-dependent electrical properties of threading dislocations in 4H-SiC

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## ABSTRACT

Electron beam induced current experiments are employed to quantitatively characterize the local electronic properties of threading dislocations (TDs) in 4H-SiC epitaxial layers. Combining with the Shockley–Read–Hall recombination theory, the local dopant concentrations and local Fermi levels of TDs are evaluated. At room temperature, the local Fermi levels of threading screw dislocations (TSDs) and threading edge dislocations (TEDs) are 14 and 29 meV higher than that of perfect region of *n*-type 4H-SiC with concentration of $4.76 \times 10^{16}\,\mathrm{cm}^{-3}$. As the doping concentration decreases to $1.83 \times 10^{14}\,\mathrm{cm}^{-3}$, the Fermi-energy difference between TSDs/TEDs and perfect region decrease to 1/18 meV, respectively. This is caused by the nitrogen accumulation at TD cores in *n*-type 4H-SiC, with the effect being more prominent for TEDs. Whereas, in *p*-type 4H-SiC, the local Fermi levels of TSDs and TEDs are 11 and 1 meV higher than those of the perfect region of 4H-SiC with a doping concentration of $3.08 \times 10^{16}\,\mathrm{cm}^{-3}$. As the doping concentration decreases to $2.55 \times 10^{15}\,\mathrm{cm}^{-3}$, the Fermi-energy differences between TSDs/TEDs and perfect 4H-SiC region increase to 22/3 meV, respectively. In *p*-type 4H-SiC, aluminum tends to be incorporated in perfect lattice sites, which is more prominent for TSDs. As the doping concentration decreases, the Fermi-energy difference decreases in *n*-type 4H-SiC, but increases in *p*-type 4H-SiC. The effect of dopants on the electronic properties of TDs increases with the increase in temperature, indicating that the adverse effect of TDs during the application of power devices based on 4H-SiC aggravates.

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Advances in 4H-SiC homoepitaxy have rendered 4H-SiC a great success in applications such as photovoltaics, electronic vehicles, rail transits, and smart grids.[1–3] Defect control, especially dislocation control, remains a central challenge in 4H-SiC epitaxy.[4,5] After decades of development, 100% conversion of basal plane dislocations (BPDs) to threading edge dislocations (TEDs) has been achieved through the design of buffer layers, promotion of lateral epitaxy along etch pits, and epitaxy suspension techniques.[6] This alleviates BPD-induced bipolar degradation.[7] However, over 95% of threading dislocations (TDs) in 4H-SiC substrates tend to replicate into epilayers.[8] Therefore, understanding the electronic properties of TDs is key to the further development of 4H-SiC materials and power devices. Thus, clarifying TD electronic properties remains essential for 4H-SiC materials and power devices.

Researchers have extensively investigated the effect of TDs on the performance of 4H-SiC. By combining *I*–*V* measurements and molten-alkali etching, threading screw dislocations (TSDs) and TEDs are found to increase the leakage current of 4H-SiC-based power devices, with TSDs playing a dominant role.[9] Photoluminescence (PL) measurements show that TDs quench band-edge emission and introduce near-infrared emission.[10,11] However, low-temperature PL is required to determine the exact defect levels of TDs, but it is difficult to precisely locate TDs under low temperature.[12] Kelvin probe force microscopy (KPFM) has also been used to investigate the local Fermi levels of dislocations, in combination with molten-alkali etching to locate TDs.[13,14] Although KPFM measurements qualitatively clarify the local Fermi levels of TDs, it is difficult to distinguish


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the effects of etching-induced defects, surface oxides, and TDs themselves. Electron beam induced current (EBIC) is highly sensitive to semiconductor defects and can quantify electronic defect properties.[15,16] It generates electron–hole pairs using a focused electron beam.[17] Their generation, diffusion, and recombination reveal carrier trapping, recombination, and transport.[18]

In this Letter, the electronic properties of TDs and their interaction with dopants in 4H-SiC are quantitatively investigated using EBIC. The higher EBIC contrast of TSDs compared to TEDs is attributed to their larger Burgers vector. Combined with Shockley–Read–Hall (SRH) theory, local doping concentrations and Fermi levels of TDs are evaluated. As the doping concentration decreases, the Fermi-level differences between TSDs/TEDs and the perfect region decrease in *n*-type 4H-SiC but increase in *p*-type 4H-SiC. This contrasting behavior is explained by nitrogen accumulation at TD cores (especially TEDs) in *n*-type material, vs aluminum incorporation into perfect lattice sites (especially for TSDs) in *p*-type material. The dopant effect also intensifies at high temperature, implying more severe TD-related degradation during high-temperature device operation.

4H-SiC homoepitaxy was carried out on 4° off-axis $n^+$-substrates using a low-pressure chemical vapor deposition reactor. $H_2$, $C_2H_4$, and trichlorosilane served as carrier, C, and Si sources. $N_2$ and trimethylaluminum supplied *n*- and *p*-type doping. The doping concentrations of 4H-SiC epilayers were characterized by a mercury CV probe (Semilab MCV-530L).

EBIC measurements were conducted using a scanning electron microscope (SEM, SN-3400) equipped with a liquid-nitrogen-cooled specimen stage. The device structures of 4H-SiC samples are illustrated in Fig. 1. Ohmic contacts were formed on the backside of the $n^+$ substrates by depositing a 200 nm nickel (Ni) layer, followed by rapid thermal annealing (RTA) at 1000 °C in an $N_2$ atmosphere. Schottky contacts on the surface of the *n*-type 4H-SiC epilayers were fabricated by depositing a 10 nm Ni layer. For the *p*-type 4H-SiC epilayers, Ohmic contacts were formed by depositing a 10 nm Ti/Al metal stack.[19] During EBIC measurements, the accelerating voltage and the beam current were set as 30 kV and 40 pA, respectively.

Molten KOH etching was adopted to confirm the existence of TDs revealed by EBIC contrasts. The 4H-SiC samples were placed in a Ni wire mesh and etched by molten KOH at 510 °C for 10 min. The 2D and 3D morphologies of the etched samples were characterized by using an optical microscope and a laser scanning confocal microscope (LSCM, ZEISS LSM900) at the same region where the EBIC experiments were performed.

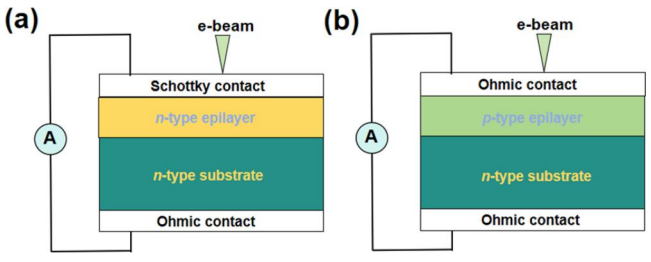



**FIG. 1.** Schematic diagrams showing device structures of (a) *n*-type and (b) *p*-type 4H-SiC epilayers for EBIC measurements.

Figure 2(a) displays the EBIC image of an *n*-type 4H-SiC epilayer with donor concentration of $4.76 \times 10^{16}\,\mathrm{cm}^{-3}$. TDs appear as dark dots, confirming them as recombination centers.[20] Based on the Burgers vectors, larger darker dots are assigned to TSDs and smaller lighter dots are assigned to TEDs. To verify the TD types identified by EBIC, molten KOH etching was performed. In Fig. 2(b), larger and smaller etch pits match the corresponding EBIC dots. Figure 2(c) compares etch-pit depth profiles with EBIC current profiles. TSD pits are deeper and wider than TED pits [Fig. 2(c)]. This agrees with stronger disorder around the larger TSD Burgers vector.[19] For both dislocation types, EBIC current decreases toward the core as atomic displacement increases. The lower TSD current indicates stronger atomic displacement and carrier recombination.[14] The larger lateral EBIC variation of TSDs also suggests a larger carrier-capture cross section [Fig. 2(c)].

The EBIC contrast ($C$), which reflects the carrier recombination character, is defined as

$$C = \frac{I_0 - I_d}{I_0}, \tag{1}$$

where $I_0$ and $I_d$ denote the EBIC currents of the perfect and TD regions of 4H-SiC, respectively. Using Eq. (1), the EBIC contrasts of TSD and TED are 8.67% and 6.00%, respectively, indicating the higher carrier recombination of TSDs.[20,21]

Temperature-dependent EBIC contrasts in differently doped 4H-SiC are then carried out to investigate dopant-TD interactions. TSD contrast remains higher than TED contrast because of stronger atomic displacement. With increasing temperature, both contrasts increase, indicating enhanced carrier-capture effect.[18,22] Lower *n*-type doping also increases both contrasts, showing stronger TD recombination at lower carrier concentration. *p*-type 4H-SiC shows stronger TD recombination than *n*-type 4H-SiC. Temperature and doping effects are also stronger in *p*-type material [Fig. 3(b)].

The SRH recombination theory is combined with EBIC measurements to quantitatively investigate the electronic properties of TDs.[22,23] TDs create continuous deep states in the bandgap of 4H-SiC.[13] In the present EBIC + SRH analysis, the threading dislocation is treated using an effective single SRH level. This level is understood as an averaged and dominant recombination level. Since the density of states of TDs exhibit Gaussian distribution, the recombination level is fitted by the empirical mode, which is defined as the maximum of the kernel density estimate as the central location during the SRH analysis.

The EBIC contrast is related to the carrier lifetime ($\tau$) by

$$C \propto \frac{1}{D\tau}. \tag{2}$$

Here, $D = kT\mu/e$ is the carrier's diffusion constant, with $k$ being the Boltzmann constant, $T$ being the temperature, $e$ being the electron charge, and $\mu$ being the minority carrier mobility. At 300 K, the minority carrier mobilities are $90\,\mathrm{cm}^2/\mathrm{V{\cdot}s}$ for *n*-type 4H-SiC and $215\,\mathrm{cm}^2/\mathrm{V{\cdot}s}$ for *p*-type 4H-SiC.[24,25] Consistent with previous research,[26] $\mu$ is assumed to be proportional to $T^{-3/2}$. Therefore, $D$ is related to $T$ by $D \propto T^{-1/2}$. First-principles calculations indicate that dislocations introduce fully occupied and empty states above the

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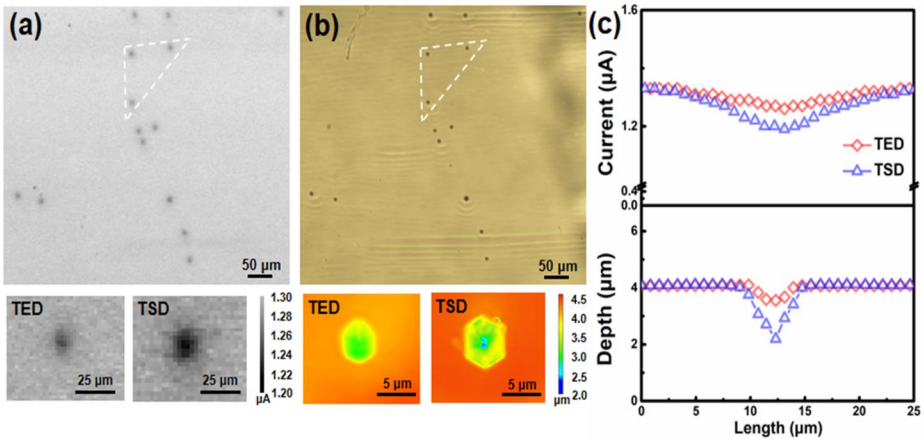



FIG. 2. (a) EBIC current images and (b) optical microscope images of etched 4H-SiC containing TDs, taken from the same perspective. The white dashed lines are used to mark the same region. The insets below panels (a) and (b) display magnified EBIC current images of TDs and LSCM images of etched TDs. (c) EBIC current profiles of TDs and depth profiles of the corresponding etch pits of TDs.

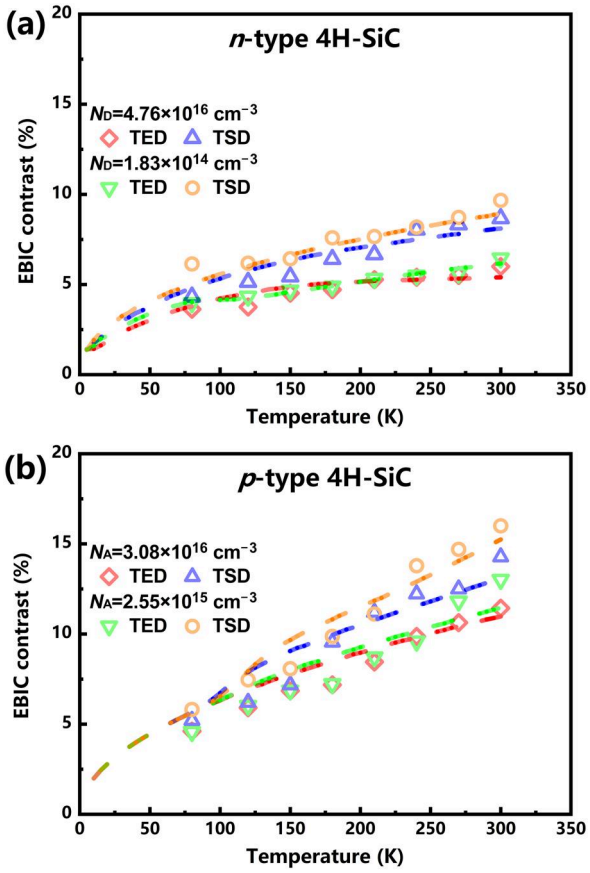



FIG. 3. Temperature dependence of EBIC contrast for TDs in (a) *n*-type 4H-SiC and (b) *p*-type 4H-SiC. Dashed curves represent fits based on SRH recombination theory. $N_D$ and $N_A$ denote the doping concentrations for *n*-type and *p*-type 4H-SiC, respectively.

valence band and below the conduction band, respectively.[27–29] The carrier capturing of dislocations are realized by electron capture via empty states and hole capture via fully occupied states. The positions of the fully occupied and empty states exert much more significant effect on the electron and hole capture process, when compared with that of the Coulomb effect. Therefore, in this work, the electron capture characteristic lifetime of electrons and holes are treated equally, and concentrate on the effect of defect-level position on the recombination behavior.

Assuming that the electron capture characteristic lifetime ($\tau_{n0}$) equals the hole capture characteristic lifetime ($\tau_{p0}$)($\tau_{n0} = \tau_{p0} = \tau_0$),[18] the carrier lifetimes ($\tau$) of TDs in *n*-type and *p*-type 4H-SiC can be described as follows:

$$\begin{aligned} \tau &= \tau_0 \frac{1 + 2\beta + n_1/n_0 + p_1/n_0}{1 + \beta}, \quad \text{for } n\text{-type 4H-SiC}, \\ \tau &= \tau_0 \frac{1 + 2\beta + n_1/p_0 + p_1/p_0}{1 + \beta}, \quad \text{for } p\text{-type 4H-SiC}. \end{aligned} \tag{3}$$

Here, $\beta$ is the injection level,[30,31] representing the ratio of the excess carrier concentration to the carrier concentration. The excess carrier concentration $\Delta p$ and $\Delta n$ can be calculated by

$$\begin{aligned} \Delta p &\approx \frac{G}{2\pi D R_e}, \quad \text{for } n\text{-type 4H-SiC}, \\ \Delta n &\approx \frac{G}{2\pi D R_e}, \quad \text{for } p\text{-type 4H-SiC}. \end{aligned} \tag{4}$$

Here, $G$ is the generation rate of electron–hole pairs excited by the electron beam per unit volume,[17] given by $G = \frac{E_b I_b}{E_{th} e}(1 - f)$. $G = \frac{E_b I_b}{E_{th} e}(1 - f)$ $E_b$ and $I_b$ are the accelerating voltage and the beam current, respectively. $E_{th}$ represents the mean energy required to create one electron–hole pair, given by $E_{th} = 2.1E_g + 1.3$,[20] where $E_g$ is the bandgap of 4H-SiC (3.26 eV). $f$ denotes the fractional electron beam energy loss due to backscattering of incident electrons, which is 0.07 for 4H-SiC at all accelerating voltages. $R_e$ is the penetration depth of the primary electron beam,[32] given by $R_e(\mu\text{m}) = 0.018\,25 E_b^{1.75}(\text{keV})$. Because the EBIC experiments were performed in the temperature range of 80 to 300 K, all relevant parameters were substituted into the formula for $\beta$, and the calculations show $\beta \ll 1$ in both *n*-type and *p*-type 4H-SiC, indicating a low injection condition.

$n_0$ and $p_0$ are the equilibrium concentrations of electrons and holes, respectively, while $n_1$ and $p_1$ denote the electron and hole concentrations at TD cores, respectively. These concentrations can be calculated by

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$$
\begin{aligned}
n_0 &= N_C \exp\left(-\frac{E_C - E_f}{kT}\right),\\
p_0 &= N_V \exp\left(-\frac{E_f - E_V}{kT}\right),\\
n_1 &= N_C \exp\left(-\frac{E_C - E_F}{kT}\right),\\
p_1 &= N_V \exp\left(-\frac{E_F - E_V}{kT}\right).
\end{aligned} \tag{5}
$$

Here, $E_C$ and $E_V$ are the energies of conduction band and valence band, respectively. With the bandgap energy of 4H-SiC being 3.26 eV, $E_C$ and $E_V$ are 3.26 and 0 eV, respectively. $N_C$ and $N_V$ are the effective density of states in the conduction and valence band, respectively, given by $N_C = 2\left(\frac{2\pi m_e^* kT}{h^2}\right)^{3/2}$ and $N_V = 2\left(\frac{2\pi m_h^* kT}{h^2}\right)^{3/2}$. Here, $m_e^*$ and $m_h^*$ are the effective masses of electron and holes, being $0.42m_0$ and $0.66m_0$, respectively.[33,34] $h$ is the Planck constant.

$E_f$ is the value of Fermi level of the perfect region, and $E_F$ is the local Fermi level at TD cores. $E_F$ is calculated by solving the following equation, with the same method applied to $E_f$,

$$
\begin{aligned}
N_C \exp\left(-\frac{E_C - E_F}{kT}\right) &= N_V \exp\left(-\frac{E_F - E_V}{kT}\right) + \frac{N_D}{1 + 2\exp\left(-\frac{E_D - E_F}{kT}\right)},\\
&\text{for } n\text{-type 4H-SiC},\\
N_V \exp\left(-\frac{E_F - E_V}{kT}\right) &= N_C \exp\left(-\frac{E_C - E_F}{kT}\right) + \frac{N_A}{1 + 4\exp\left(-\frac{E_F - E_A}{kT}\right)},\\
&\text{for } p\text{-type 4H-SiC}.
\end{aligned} \tag{6}
$$

Here, $E_D$ and $E_A$ represent the ionization energies of N and Al at TD cores for $n$-type and $p$-type 4H-SiC, respectively, with values of 61.8 and 200 meV, respectively. We note that the strain field induced by dislocations would affect the band edges of 4H-SiC. When the local strain field associated with a dislocation shifts the conduction and valence band edges upward or downward, the N donor level and the Al acceptor level are expected to move together with their corresponding band edges.[35] Therefore, the donor ionization energy measured from the CBM and the acceptor ionization energy measured from the valence band maximum (VBM) are not expected to change as strongly as the absolute band-edge energies.

Using Eqs. (2)–(6), the EBIC contrast can be expressed by introducing a constant $\alpha$ as follows:

$$
\begin{aligned}
C &= \frac{\alpha T^{1/2}(1+\beta)}{1 + 2\beta + n_1/n_0 + p_1/n_0}, \quad \text{for } n\text{-type 4H-SiC},\\
C &= \frac{\alpha T^{1/2}(1+\beta)}{1 + 2\beta + n_1/p_0 + p_1/p_0}, \quad \text{for } p\text{-type 4H-SiC}.
\end{aligned} \tag{7}
$$

Fitting Eq. (7) to EBIC contrast gives the local carrier concentrations of TDs. The values of the fitting parameters $N_D$, $N_A$ and the fitting accuracies ($R^2$) are tabulated in Table I. The high accuracies (0.82–0.96) indicate reliable fitting. For $n$-type 4H-SiC with a doping concentration of $4.76 \times 10^{16}\,\mathrm{cm}^{-3}$, the local carrier concentrations at the cores of TSD and TED are $8.65 \times 10^{16}$ and $1.81 \times 10^{17}\,\mathrm{cm}^{-3}$, respectively. When the doping concentration decreases to $1.83 \times 10^{14}\,\mathrm{cm}^{-3}$, the local carrier concentrations at the cores of TSD and TED are $1.94 \times 10^{14}$ and $3.87 \times 10^{14}\,\mathrm{cm}^{-3}$, respectively. Thus, TD cores have higher carrier concentrations than the perfect $n$-type region, especially for TEDs.

For $p$-type 4H-SiC, with a doping concentration of $3.08 \times 10^{16}\,\mathrm{cm}^{-3}$, the local carrier concentrations at the cores of TSD and TED are $1.48 \times 10^{16}$ and $2.96 \times 10^{16}\,\mathrm{cm}^{-3}$, respectively. When the doping concentration decreases to $2.55 \times 10^{15}\,\mathrm{cm}^{-3}$, the local carrier concentrations at the cores of TSD and TED decrease to $8.03 \times 10^{14}$ and $2.22 \times 10^{15}\,\mathrm{cm}^{-3}$, respectively. Thus, TD cores have lower carrier concentrations than the perfect $p$-type region, especially for TSDs.

With the local carrier concentration at TD cores, we calculate the local Fermi levels at TD cores in differently doped 4H-SiC. Under the preconditions of low-injection, where $n_0 \gg p_1$ for $n$-type 4H-SiC and $p_0 \gg n_1$ for $p$-type 4H-SiC, Eq. (7) can be simplified as

$$
\begin{aligned}
C &= \frac{\alpha T^{1/2}}{1 + \exp\left(\frac{E_F - E_f}{kT}\right)}, \quad \text{for } n\text{-type 4H-SiC},\\
C &= \frac{\alpha T^{1/2}}{1 + \exp\left(-\frac{E_F - E_f}{kT}\right)}, \quad \text{for } p\text{-type 4H-SiC}.
\end{aligned} \tag{8}
$$

Using Eq. (8), the fitting parameters are essentially the same as those obtained from Eq. (7). The calculated local Fermi levels at TSD and TED cores in differently doped 4H-SiC are displayed in Fig. 4.

For $n$-type 4H-SiC with a doping concentration of $4.76 \times 10^{16}\,\mathrm{cm}^{-3}$, the local Fermi levels of TED and TSD cores lie at

**TABLE I.** Fitting results of EBIC contrasts for TDs in differently doped $n$-type and $p$-type 4H-SiC.

| $n$-type 4H-SiC | | | | $p$-type 4H-SiC | | | |
|---|---|---|---|---|---|---|---|
| Doping concentration ($\mathrm{cm}^{-3}$) | TD type | $N_D$ ($\mathrm{cm}^{-3}$) | $R^2$ | Doping concentration ($\mathrm{cm}^{-3}$) | TD type | $N_A$ ($\mathrm{cm}^{-3}$) | $R^2$ |
| $4.76 \times 10^{16}$ | TSD | $8.65 \times 10^{16}$ | 0.85 | $3.08 \times 10^{16}$ | TSD | $1.48 \times 10^{16}$ | 0.88 |
| $4.76 \times 10^{16}$ | TED | $1.81 \times 10^{17}$ | 0.83 | $3.08 \times 10^{16}$ | TED | $2.96 \times 10^{16}$ | 0.86 |
| $1.83 \times 10^{14}$ | TSD | $1.94 \times 10^{14}$ | 0.82 | $2.55 \times 10^{15}$ | TSD | $8.03 \times 10^{14}$ | 0.94 |
| $1.83 \times 10^{14}$ | TED | $3.87 \times 10^{14}$ | 0.96 | $2.55 \times 10^{15}$ | TED | $2.22 \times 10^{15}$ | 0.83 |

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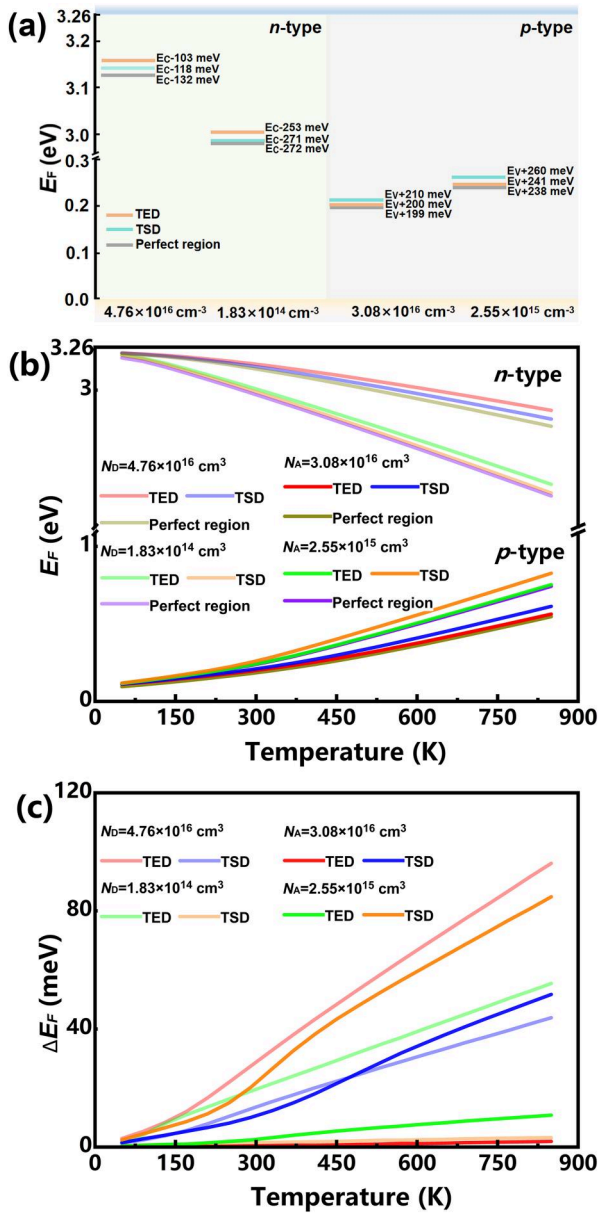



**FIG. 4.** (a) Positions of local Fermi levels ($E_F$) of TDs in *n*-type and *p*-type 4H-SiC at room temperature. The Fermi levels are proportionally labeled with a 300 meV interval between the band edge and the mid-gap. (b) Temperature dependence of the Fermi level of TDs and perfect regions for *n*-type and *p*-type 4H-SiC. (c) Temperature dependence of the Fermi-energy difference ($\Delta E_F$) between TDs and the perfect regions.

103 and 118 meV below the CBM, respectively, which are higher than that in the perfect region by 29 and 14 meV, respectively. When the doping concentration decreases to $1.83 \times 10^{14}\,\mathrm{cm}^{-3}$, the local Fermi levels of TED and TSD cores shift downward to 253 and 271 meV below the CBM, respectively, and they are higher than those in the perfect region by 18 and 1 meV, respectively. Thus, TDs act as shallow donors in *n*-type 4H-SiC, with TSD levels deeper than TED levels. When the doping concentration decreases, the perfect-region Fermi level shifts deeper as free-electron concentration decreases. The TD local Fermi energies shift more strongly, especially for TSDs.

For *p*-type 4H-SiC with a doping concentration of $3.08 \times 10^{16}\,\mathrm{cm}^{-3}$, the local Fermi levels of TED and TSD cores lie at 200 and 210 meV above the VBM of 4H-SiC, respectively, which are higher than that in the perfect region by 1 and 11 meV, respectively. When the doping concentration decreases to $2.55 \times 10^{15}\,\mathrm{cm}^{-3}$, the local Fermi levels of TED and TSD cores shift upward to 241 and 260 meV above the VBM, respectively, and they are higher than those in the perfect region by 3 and 22 meV, respectively. Thus, TDs act as deep acceptors in *p*-type 4H-SiC, with TSD levels deeper than TED levels. When the doping concentration decreases, the perfect-region Fermi level shifts deeper as free-hole concentration decreases. Again, TD local Fermi levels shift more strongly, especially for TSDs.

Since 4H-SiC is attractive for high-temperature power electronics,[36] we then quantitatively evaluate the effect of temperature on the local Fermi levels of TDs and the perfect region, as well as the Fermi-energy difference ($\Delta E_F$) between them in differently doped 4H-SiC, as shown in Fig. 4(b) and 4(c). For *n*-type 4H-SiC, TD and perfect-region Fermi levels are nearly identical at low temperature. With increasing temperature, both TD and perfect-region Fermi levels shift deeper. However, TDs, especially TEDs, shift less than the perfect region. Thus, N-related regulation strengthens with temperature, particularly for TEDs. When the doping concentration decreases, TD Fermi levels, especially for TSDs, approach the perfect-region level. At a doping concentration of $4.76 \times 10^{16}\,\mathrm{cm}^{-3}$ and a temperature of 50 K, the $\Delta E_F$ of TSDs and TEDs are 1 and 3 meV, respectively, while at a doping concentration of $1.83 \times 10^{14}\,\mathrm{cm}^{-3}$, the corresponding values are 0 and 2 meV, respectively. When the temperature increases to 850 K, the $\Delta E_F$ of TSDs and TEDs increase to 44 and 96 meV, respectively, at a doping concentration of $4.76 \times 10^{16}\,\mathrm{cm}^{-3}$, and to 3 and 55 meV, respectively, at a doping concentration of $1.83 \times 10^{14}\,\mathrm{cm}^{-3}$. Elevated temperature therefore enhances electron transfer between N atoms and dislocation states. This effect is stronger for TEDs and weakens with decreasing doping.

For *p*-type 4H-SiC, TD and perfect-region Fermi levels are also nearly identical at low temperature. When the temperature increases, both shift deeper, but TDs, especially TSDs, shift more than the perfect region. Thus, elevated temperature deepens TD levels through stronger lattice distortion, mainly for TSDs. When the doping concentration decreases, TD Fermi levels, especially for TSDs, move farther from the perfect-region level. The $\Delta E_F$ of TSDs and TEDs are 2 and 0 meV, respectively, for a doping concentration of $3.08 \times 10^{16}\,\mathrm{cm}^{-3}$ at 50 K, and the corresponding values are 3 and 1 meV, respectively, for a doping concentration of $2.55 \times 10^{15}\,\mathrm{cm}^{-3}$. When the temperature increases to 850 K, the $\Delta E_F$ of TSDs and TEDs increase to 52 and 2 meV, respectively, at a doping concentration of $3.08 \times 10^{16}\,\mathrm{cm}^{-3}$, and to 85 and 11 meV, respectively, at a doping concentration of $2.55 \times 10^{15}\,\mathrm{cm}^{-3}$. The effect is strongest for TSDs and increases as doping decreases.

The mechanism follows from dopant-dislocation interactions in 4H-SiC. The local Fermi levels of TEDs and TSDs are higher than those of the perfect region in both doping types. Thus, TDs are shallower in *n*-type 4H-SiC but deeper in *p*-type 4H-SiC relative to the perfect region. Known BPD-dopant interactions help to explain the

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TD behavior. In *n*-type 4H-SiC, electron transfer from N defect states to dislocation states lowers system energies and drives N accumulation at the cores.[29] This N enrichment explains the shallower TD Fermi levels. In *p*-type material, symmetry forbids electronic interaction between Al and dislocation defect states.[37] Al aggregation at cores would worsen lattice distortion, lowering Al incorporation at the cores.[14] This Al depletion explains the deeper TD Fermi levels in *p*-type 4H-SiC. The dopant effect strengthens with temperature, aggravating TD-related degradation in 4H-SiC power devices.

In summary, EBIC measurements are used to quantify the doping-dependent electronic properties of TDs in 4H-SiC. TSDs and TEDs appear as larger and smaller dark spots, respectively, confirming TD's recombination activity. Higher TSD contrast arises from its larger Burgers vector and stronger local atomic displacement. Combining the EBIC current with SRH theory yields local TD doping concentrations and Fermi levels. TD local Fermi levels are higher than those of perfect regions in both doping types, but with different mechanisms. In *n*-type 4H-SiC, the Fermi-energy difference between TED and the perfect region exceeds the TSD difference and decreases with decreasing doping. This originates from N accumulation at TD cores, especially TEDs. In *p*-type 4H-SiC, Al preferentially incorporates into perfect regions, with stronger Al repulsion for TSDs. The dopant effect increases with temperature, so TD-related degradation becomes more severe during high-temperature operations. These results support quantitative assessment of TD effects on 4H-SiC device reliability.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

**Zedong Tong:** Data curation (lead); Methodology (lead); Writing – original draft (lead); Writing – review & editing (lead). **Jifa Liang:** Investigation (equal). **Xiaowei Zhang:** Formal analysis (equal). **Xiangxiang Zhao:** Formal analysis (equal). **Haibing Zhang:** Investigation (equal). **Guixiu Li:** Investigation (equal). **Lei Yang:** Investigation (equal). **Shuai Yuan:** Conceptualization (equal). **Can Cui:** Writing – review & editing (equal). **Xiaodong Pi:** Resources (equal). **Deren Yang:** Resources (equal); Supervision (equal). **Rong Wang:** Conceptualization (lead); Funding acquisition (lead); Supervision (lead); Writing – review & editing (lead).

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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